Amulet™ Steerable Delivery Sheath

Instructions for Use

CAUTION: Read all instructions carefully. Failure to follow the instructions, warnings, and precautions may lead to patient injury or device damage.

Device Description

The Amulet™ Steerable Delivery Sheath (STRBL-14F-075) is designed to provide a pathway through which a device may be delivered. The sheath is provided in one size (14Fr), with a working length of 75 cm and a bidirectional distal tip to provide positioning in the cardiac anatomy. The handle is equipped with a deflection knob to deflect the tip clockwise 120° and counterclockwise 0°. The body of the sheath is radiopaque for visibility under fluoroscopy and has a marker band located in the distal tip. The dilator eases penetration of tissue. The sheath and dilator utilize a dual curve in two dimensions, resulting in a three-dimensional geometry. The 14Fr flush adapter facilitates the attachment of additional components.

See Table 1 for a compatibility chart; see Table 2 for delivery sheath dimensions. Figure 1 through Figure 3 illustrate the following components:

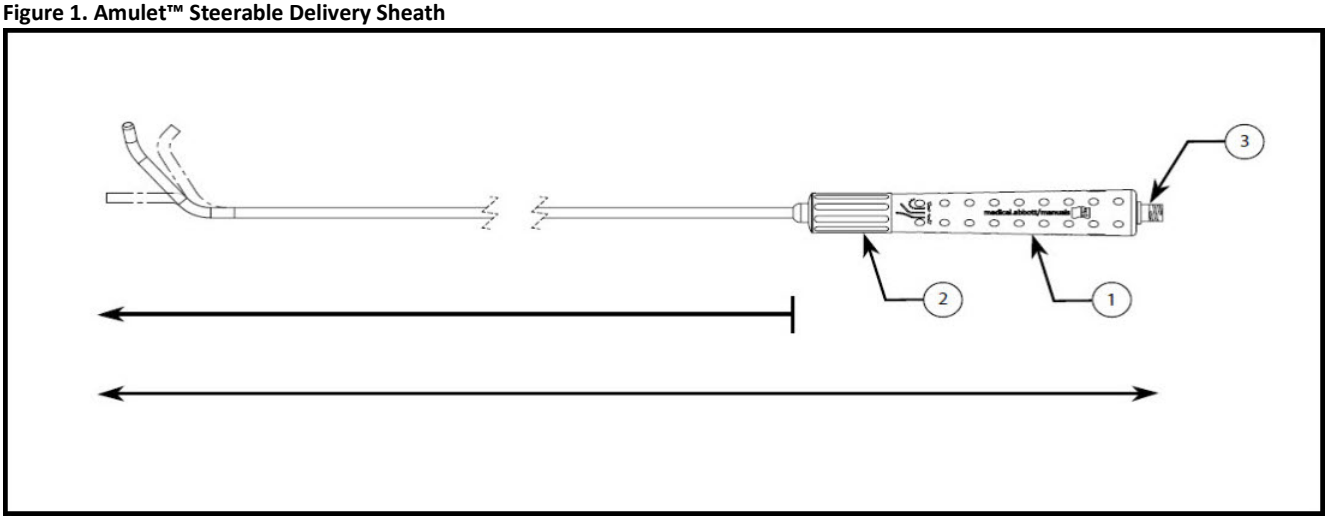
Figure 1 Delivery Sheath

1. Handle
2. Handle deflection knob
3. Sheath hub

Figure 2 Dilator

Figure 3 14Fr Flush Adapter

Figure 1. Amulet™ Steerable Delivery Sheath



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Figure 2. Amulet™ Steerable Delivery Sheath Dilator

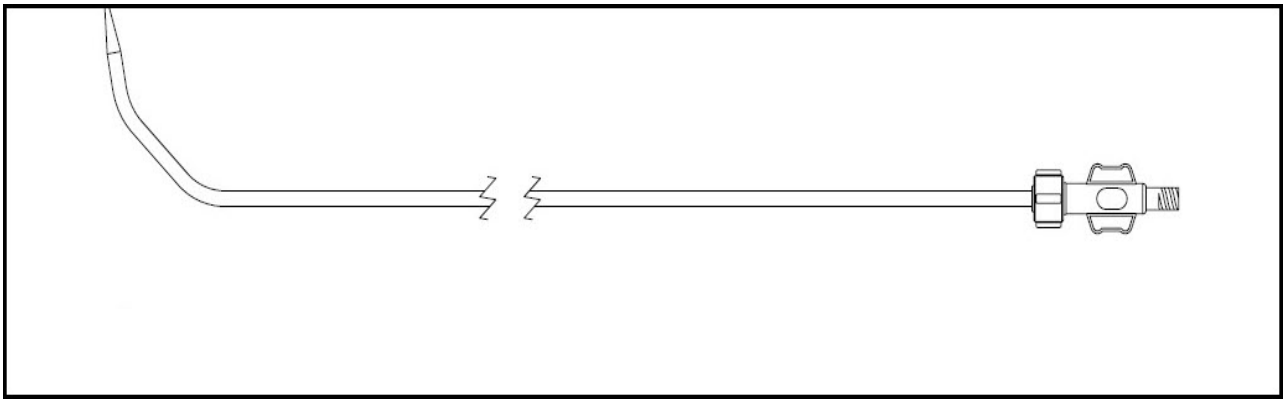


Figure 3. 14Fr Flush Adapter

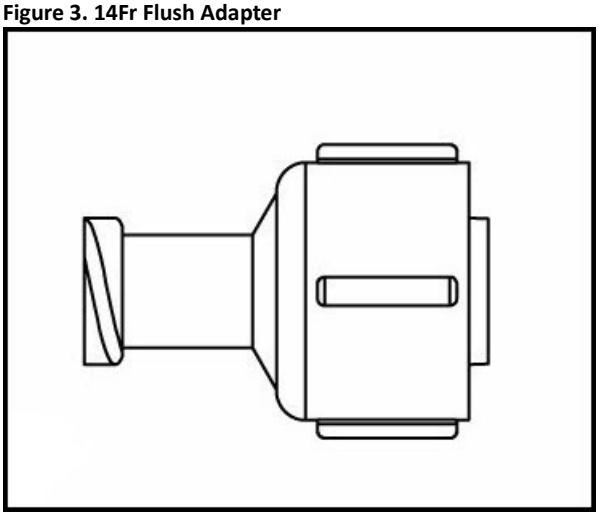


Table 1. Compatibility chart

Product Description	Amplatzer™ Amulet™ Size (mm)	Model Numbers
Amplatzer™ Amulet™ Left Atrial Appendage Occluder	16	9-ACP2-007-016
	18	9-ACP2-007-018
	20	9-ACP2-007-020
	22	9-ACP2-007-022
	25	9-ACP2-010-025
	28	9-ACP2-010-028
	31	9-ACP2-010-031
	34	9-ACP2-010-034

NOTE: The Amulet™ Steerable Delivery Sheath has not been tested for use with other devices.

Table 2. Amulet™ Steerable Delivery Sheath dimensions

REF					
	Fr	mm (in)	mm (in)	cm (in)	cm (in)
STRBL-14F-075	14	4.7 (0.184)	6.0 (0.238)	75.0 (29.5)	93.9 (37.0)

Indication for Use

The Amulet™ Steerable Delivery Sheath is indicated to facilitate the delivery of the Amplatzer™ Amulet™ Left Atrial Appendage Occluder.

Contraindications

The Amulet™ Steerable Delivery Sheath is contraindicated for patients:

- with the presence of thrombus that is in the pathway of the steerable sheath, including within the left atrial appendage
- with active infections including endocarditis

The Amulet™ Steerable Delivery Sheath is contraindicated for use with a power injection system.

Warnings

- This device was sterilized with ethylene oxide and is for single use only. Do not reuse or resterilize this device. Attempts to resterilize this device can cause a malfunction, insufficient sterilization, or harm to the patient.
- Do not use this device if the sterile package is open or damaged. Inspect all components before use. Do not use if the package or items appear to be damaged or defective.
- This device should be used only by physicians who are trained in standard transcatheter techniques. The physician should determine which patients are candidates for procedures that use this device.
- Slowly remove the dilator and sheath from the patient to prevent an ingress of air.
- As this medical device uses an alternative small-bore connector (luer) design different from those of the ISO 80369 series, there is a possibility that a misconnection can occur between this medical device and a medical device using a different alternative small-bore connector, which can result in a hazardous situation causing harm to the patient. Special measures need be taken by the user to mitigate these reasonable, foreseeable risks. See call-out 3 in Figure 1 for an illustration of the small-bore connector (luer).
- Failure to cap the sheath hub once the dilator has been removed may result in excessive bleeding or air ingress.
- When removing the delivery sheath from the patient, failure to return the delivery sheath to a neutral position may result in vascular or cardiac injury.
- Do not deflect the Amulet™ Steerable Delivery Sheath to its maximum deflection more than 10 times during the procedure.

Precautions

- DO NOT use the Amulet™ Steerable Delivery Sheath after the Use-by date stated on the package label.
- The physician should exercise clinical judgment in situations that involve the use of anticoagulants or antiplatelet drugs before, during, or after the use of the Amulet™ Steerable Delivery Sheath.
- When placing an Amplatzer™ Amulet™ Left Atrial Appendage Occluder using the Amulet™ Steerable Delivery Sheath, refer also to the instructions for use provided with the Amplatzer™ Amulet™ Left Atrial Appendage Occluder.
- Do not over-tighten the connection between the dilator rotating luer and delivery sheath hub.
- Use caution when advancing the Amulet™ Steerable Delivery Sheath and dilator to avoid damaging tissue and vessels or interfering with previously implanted medical devices. If resistance is felt, determine the cause before proceeding. Continuing to advance, retract, or manipulate the catheter under this circumstance has the potential to result in patient injury or delivery sheath damage.

Limitations

Certain conditions may require special consideration when using this product. These may be, but are not limited to calcified or thickened interatrial septum, or cases with previously implanted medical devices (for example, a septal occluder or transvenous pacing leads)

Potential Adverse Events

Potential adverse events that can occur during or after a procedure using this sheath may include but are not limited to:

- Air embolism
- Allergic reaction/toxic effects (due to anesthesia, contrast media, etc.)
- Arrhythmia
- Arteriovenous fistulae
- Bleeding
- Cardiac tamponade
- Cardiac perforation
- Death
- Dissection
- Embolism
- Foreign material embolus
- Heart failure
- Hematoma
- Hypotension/hypertension
- Infection
- Myocardial infarction/ischemia
- Pericardial effusion
- Peripheral embolism
- Pseudoaneurysm
- Stroke
- Thrombosis
- Transient ischemic attack
- Vascular access site injury (such as arteriovenous fistulae, hematoma, pseudoaneurysm)
- Vessel trauma/damage

Reporting Device Incidents

If, in the course of use of this device, you have reason to believe that a serious incident occurred, please report it to the manufacturer. For user facilities in the United States, report a medical device-related serious injury to the manufacturer, or to the FDA if the medical device manufacturer is unknown.

Device Compatibility

This device is compatible with the Amplatzer™ Amulet™ loader. A standard hypodermic syringe may be attached to the sheath hub by first attaching the 14Fr flush adapter, and then connecting the syringe to the flush adapter.

Performance Characteristics

The Amulet™ Steerable Delivery Sheath consists of the following:

- Steerable Delivery Sheath
- Dilator
- 14Fr flush adapter

With a working length of 75 cm and an inner diameter that is 14Fr, the Amulet™ Steerable Delivery Sheath provides flexible catheter positioning in the cardiac anatomy. The delivery sheath handle is equipped with a handle deflection knob to deflect the tip bidirectionally. The pre-set bidirectional curve aligns with typical left atrial appendage anatomy (see Figure 1 and Figure 2). The delivery sheath includes a radiopaque tip marker to improve fluoroscopic visualization.

Storage and Handling Information

- There are no special storage requirements beyond what is typical for medical devices.
- The instructions for use is recyclable. Dispose of all packaging materials appropriately.

Directions for Use

Materials recommended for use with the Amulet™ Steerable Delivery Sheath:

- 0.035-inch guidewire
- 19Fr or larger introducer sheath
- 30cc-60cc luer-lock syringe
- 5Fr - 6Fr marker pigtail catheter, 110 cm length minimum
- Hemostasis valve

Procedure

CAUTION: When placing an Amplatzer™ Amulet™ Left Atrial Appendage Occluder using the Amulet™ Steerable Delivery Sheath, refer also to the instructions for use provided with the device.

1. Prepare the Amulet™ Steerable Delivery Sheath for use:
   - Inspect the sterile pouch and verify that it is unopened and undamaged. Do not use the components if the sterile barrier has been compromised.
   - Gently open the sterile pouch and inspect the contents for damage. Do not use damaged or kinked components.
   - Ensure the delivery sheath curve can be easily deflected. Do not use the delivery sheath if there is difficulty in deflecting the distal section to achieve or hold the desired curve.
   - Flush the components with sterile saline.

   NOTE: Connect the 14Fr flush adapter to the sheath hub before connecting a syringe to flush.

   - Wipe the delivery sheath and dilator with sterile gauze dampened with sterile saline.
2. Insert the dilator into the delivery sheath and turn the rotating luer on the dilator clockwise to lock the components together.

   NOTE: You may encounter resistance as the dilator reaches the distal end of the delivery sheath because the tip of the sheath is tapered.

   CAUTION: Do not over-tighten the connection.
3. Access the left atrium using transvenous and transseptal techniques and equipment.

   NOTE: After transseptal puncture is performed, maintain the recommended activated clotting time (ACT) of 250 to 350 seconds until the procedure is complete.
4. Place a 0.035-inch guidewire with the distal end in the left upper pulmonary vein.
5. Advance the delivery sheath and dilator over the guidewire until the delivery sheath is positioned according to the Amplatzer™ Amulet™ Left Atrial Appendage Occluder instructions.

   CAUTION: If resistance is encountered, use available imaging modalities to determine the cause before proceeding. Use of excessive force may result in patient injury or delivery sheath damage. The delivery sheath may be retracted slightly and straightened prior to being re-advanced.
6. Turn the rotating luer on the dilator counter-clockwise to unlock the components. Slowly remove the dilator and guidewire from the delivery sheath.

   WARNING: Remove the dilator and guidewire slowly to prevent air ingress.
7. To prevent blood loss and avoid air ingress, use a hemostasis valve or otherwise ensure that the sheath hub is capped until ready to connect the device loader.

   WARNING: Failure to cap the sheath hub may result in excessive bleeding or air embolism.

   WARNING: The sheath hub is a custom luer fitting and is only compatible with the supplied 14Fr flush adapter, 14Fr sheath adapter, and the device's 14Fr loader. Do not attempt to directly connect any other accessories to the sheath hub.
8. Remove the hemostasis valve or cap that was placed on the sheath hub after removal of the dilator. Allow fluid backflow to purge air from the delivery sheath by holding the sheath hub below the plane of the patient's body.
9. Immediately connect the loader of the device to the sheath in a fluid-to-fluid fashion, according to the device's instructions for use.
10. Advance the device to the tip of the delivery sheath per the device's instructions for use.

11. Adjust the tip of the delivery sheath by turning the handle deflection knob to achieve the desired position in the patient's anatomy.

    NOTE: Deflection direction is shown in graphics on the handle.

12. Implant the occluder according to the device's instructions for use.
13. When the procedure is complete, return the tip of the delivery sheath to a neutral (or atraumatic) position, and slowly remove the delivery sheath.

    WARNING: Remove the delivery sheath slowly to prevent air ingress.

    WARNING: Failure to return the delivery sheath to a neutral position may result in vascular or cardiac injury.

Post-procedure Care

Instruct the patient when to seek medical attention.

Disposal

The Amulet™ Steerable Delivery Sheath should be classified as a biohazard and disposed of in compliance with facility procedures and applicable laws and regulations.

The instructions for use are recyclable. Dispose of all packaging materials appropriately.

Warranty

Abbott Medical warrants to buyer that, for a period equal to the validated shelf life of the product, this product shall meet the product specifications established by the manufacturer when used in accordance with the manufacturer's instructions for use and shall be free from defects in materials and workmanship. Abbott Medical's obligation under this warranty is limited to replacing or repairing at its option, at its factory, this product if returned within the warranty period to Abbott Medical and after confirmed to be defective by the manufacturer.

EXCEPT AS EXPRESSLY PROVIDED IN THIS WARRANTY, ABBOTT MEDICAL DISCLAIMS ANY REPRESENTATION OR WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING ANY WARRANTY AS TO MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE.

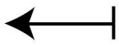
See the terms and conditions of sale for further information.

State of California (USA) Only:

⚠ **WARNING: This product can expose you to chemicals including ethylene oxide, which is known to the State of California to cause cancer and birth defects or other reproductive harm. For more information, go to www.P65Warnings.ca.gov.**

Symbol Definitions

The symbols below and harmonized symbols may be found on the product or product label. For harmonized symbols, refer to the Universal Symbols Glossary at medical.abbott/manuals.

Symbol	Definition
	Caution, consult accompanying documents
	Manufacturer
	Reference number
	Batch Code
	Use-by date
	Do not re-use
	Sterilized using ethylene oxide
	Unique device identifier
	Follow instructions for use on this website
	Keep dry; keep away from rain
	Do not use if package is damaged
	Inner diameter
	Outer diameter
	Length
	Usable length
	Amulet™ Steerable Delivery Sheath

Symbol	Definition
	Federal law restricts this device to sale by or on the order of a physician.
	Quantity
	Date of manufacture
Steerable Delivery Sheath	Steerable Delivery Sheath
	Do not resterilize

™ Indicates a trademark of the Abbott group of companies.
‡ Indicates a third party trademark, which is property of its respective owner.
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